# Theory of Value

## An Axiomatic Analysis of Economic Equilibrium
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# Chapter 1: Mathematics

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# Real Numbers

The set $\mathbb{R}$ of (finite, real) numbers is defined as a set of elements having the following properties (all familiar, perhaps with the exception of the last one).

# Real Numbers

There are on $\mathbb{R}$ two associative and commutative binary operations (addition $+$, and multiplication $\cdot$). There is an element, 0, which, if added to any element $x$, gives $x$. Any element $x$ has a negative, i.e., an element which, if added to $x$, gives 0. There is an element, 1, which if multiplied by any element $x$, gives $x$. Any element $x$ different from 0 has an inverse, i.e., an element which, if multiplied by $x$, gives 1. Multiplication is distributive with respect to addition. 0 is different from 1.

# Real Numbers

There is on $\mathbb{R}$ a complete ordering denoted $\leqslant$. If $z$ is any element and $x_1 \leqslant x_2$, then $x_1 + z \leqslant x_2 + z$. If $0 \leqslant x_1$ and $0 \leqslant x_2$, then $0 \leqslant x_1 \cdot x_2$.

# Real Numbers

Finally, every non-empty subset $X$ of $\mathbb{R}$ which has an upper bound has a least upper bound, and every non-empty subset $X$ of $\mathbb{R}$ which has a lower bound has a greatest lower bound.

# Real Numbers

A few definitions and further (derived) properties of $\mathbb{R}$ will now be added. $x_1 + x_2$ is the sum of $x_1$ and $x_2$, $x_1 \cdot x_2$ is their product. If $x_1, \cdots, x_n, \cdots, x_N$ are $N$ elements of $\mathbb{R}$, their sum is denoted $\sum_{n=1}^{N} x_n$. The product of $N$ elements equal to $x$ is denoted $x^N$ and called the $N$th power of $x$. 0 (resp. 1) is the only element having the property which defines 0 (resp. 1). Any $x \in \mathbb{R}$ has a unique negative denoted $-x$. One writes $x_1 - x_2$ for $x_1 + (-x_2)$. The corresponding binary operation is called subtraction, and the result difference. Any $x \in \mathbb{R}$ different from 0 has a unique inverse denoted $\frac{1}{x}$. One writes $\frac{x_1}{x_2}$ for $x_1 \cdot \frac{1}{x_2}$. The corresponding binary operation is called division, and the result quotient. For any $x \in \mathbb{R}$ one has $0 \cdot x = 0$, and $(-1) \cdot x = -x$. The multiplication dot will now always be dropped.

# Real Numbers

$x_1 \leqslant x_2$ (resp. $x_1 < x_2$) is read $x_1$ less than or equal to $x_2$ (resp. $x_1$ less than $x_2$). $x \leqslant 0$ (resp. $x < 0$) is read $x$ non-positive (resp. $x$ negative). The expressions are transposed in an obvious way if the inequality sign is inverted. One has $0 < 1$.

# Real Numbers

A greatest (resp. least) element of a subset $X$ of $\mathbb{R}$, if it exists, is unique; it is called the maximum (resp. minimum) of $X$ and denoted $\max X$ (resp. $\min X$). One defines the absolute value $|x|$ of a number $x$ by $|x| = \max\{x, -x\}$, and the sign of a number $x \neq 0$ by $\operatorname{sign} x = \dfrac{x}{|x|}$. One has $|x_1 + x_2| \leqslant |x_1| + |x_2|$.

# Real Numbers

A least upper bound of a subset $X$ of $\mathbb{R}$, if it exists, is unique (1.4.f); it is called the supremum of $X$ and denoted $\sup X$. Every non-empty subset $X$ of $\mathbb{R}$ which has a lower bound has a unique greatest lower bound called the infimum of $X$ and denoted $\inf X$.

The simplest example is provided by the "non-empty interval $I = [a, b)$. It has no maximum, its supremum is $b$. Its minimum, and therefore (1.4.g) its infimum, is $a$. See also example (2) of 1.5.o.

# Real Numbers

The infimum (resp. supremum) of a real interval is also called its origin (resp. extremity). The length of an interval with origin $a$ and extremity $b$ is $b - a$.

# Real Numbers

By repeated addition of 1 to 0, and repeated subtraction of 1 from 0, one obtains the set $\mathbb{Z}$ of integers (non-negative and non-positive) as a subset of $\mathbb{R}$.

# Real Numbers

A real number of the form $\frac{p}{q}$, where $p \in \mathbb{Z}$ and $q \in \mathbb{Z}$, and $q \neq 0$, is called a rational number. The set of rationals, a subset of $\mathbb{R}$, is denoted by $\mathbb{Q}$.

Theorem 1. The set $\mathbb{Q}$ is countable.

(use (2) of 1.3.n to prove that $\mathbb{Z}$, then $\mathbb{Q}$ is countable) and satisfies all the axioms in the definition of $\mathbb{R}$ with the exception of the last one. (For example, one can prove that the set $\left\{x \in \mathbb{Q} \mid x^2 < 2\right\}$ has no least upper bound in $\mathbb{Q}$, using the fact that there is no rational $y$ satisfying $y^2 = 2$.)

# Real Numbers

One can prove that the set $\mathbb{R}$ is not countable. One can also prove that

Theorem 2. If $x, y$ belong to $\mathbb{R}$ and satisfy $x < y$, there is a rational $r$ such that $x < r < y$.

# Real Numbers

Given a number $x \geqslant 0$ and a positive integer $N$, one can prove that there is a unique $y \geqslant 0$ such that $y^N = x$. One calls $y$ the $N$th root of $x$, and one writes $y = x^{\frac{1}{N}}$.

# Real Numbers

Consider a sequence $(x^q)$ of numbers. Intuitively, one says that $(x^q)$ converges (or tends) to the number $x^0$ if $x^q$ is as close to $x^0$ as one wishes provided that $q$ is large enough. In a precise fashion, $(x^q)$ converges to $x^0$ if, for any number $\varepsilon > 0$, there is an integer $q' > 0$ (depending on $\varepsilon$) such that $q > q'$ implies $\left|x^q - x^0\right| < \varepsilon$. One writes $x^q \to x^0$.

# Real Numbers

A sequence which tends to a number is called convergent. It is clear that "$x^q \to x^0$ and $x^q \to x^*$" implies "$x^0 = x^*$"; thus a convergent sequence tends to a unique number called its limit.

Example: (1) The sequence $\left(\frac{(-1)^q}{q}\right)$ tends to 0 since, given $\varepsilon > 0$, making $\frac{1}{q} < \varepsilon$ is equivalent to making $q > \frac{1}{\varepsilon}$. (2) Note that if $X$ is the set $\left\{\frac{1}{1}, \frac{1}{2}, \cdots, \frac{1}{q}, \cdots\right\}$, then $X$ has no minimum and $\inf X = 0$. (3) The sequences $((-1)^q)$ and $(q)$ are not convergent.

# Real Numbers

The elements of $\mathbb{R}$ are also called points. The set $\mathbb{R}$ may be visualized as follows. On a horizontal straight line, two different points are chosen; they will represent 0 and 1. 1 being to the right of 0. An element $x$ of $\mathbb{R}$ is then represented by a point of the straight line at distance $|x|$ from 0, to the right (resp. left) of 0 if $x$ is positive (resp. negative).

# Real Numbers

The letters $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ have throughout this volume the meanings introduced respectively in 1.2.a, 1.5.j, 1.5.k, 1.5.a.

# Real Number System

There are at least two approaches to defining the real number system.

Approach 1. Start with the set of natural numbers $\mathbb{N}$, construct the set $\mathbb{Z}$ of all integers $(\mathbb{N} \cup \{0\} \cup (-\mathbb{N}))$, and next, the set $\mathbb{Q}$ of rationals and then the set of $\mathbb{R}$ of real numbers either as the set of all Cauchy sequences of rationals or as Dedekind cuts. The step going from $\mathbb{Q}$ to $\mathbb{R}$ via Cauchy sequences is also available for completing any incomplete metric space.

# Real Number System

There are at least two approaches to defining the real number system.

Approach 2. Define the set of real numbers, $\mathbb{R}$, as a set that satisfies three sets of axioms. The first set is algebraic involving addition and multiplication. The second set is on ordering that, with the first, makes $\mathbb{R}$ an ordered field (Royden 1988). The third set is a single axiom known as the completeness axiom. Thus $\mathbb{R}$ is defined as complete ordered field.

# Additive Group

A group is a set, $G$, together with an operation $+$ (called the group law of $G$) that combines any two elements $a_1$ and $a_2$ to form another element, denoted $a_1 + a_2$. To qualify as a group, the set and operation, $(G, +)$, must satisfy four requirements known as the group axioms:

(1) Closure: For all $a_1$ and $a_2$ in $G$, the result of the operation, $a_1 + a_2$, is also in $G$.

(2) Associativity: For all $a_1$, $a_2$ and $a_3$ in $G$, $(a_1 + a_2) + a_3 = a_1 + (a_2 + a_3)$.

(3) Identity element: There exists an element 0 in $G$ such that, for every element $a$ in $G$, the equation $0 + a = a + 0 = a$ holds. Such an element is unique, and thus one speaks of the identity element.

(4) Additive Inverse: For each $a$ in $G$, there exists an inverse element $-a$ in $G$ such that $a + (-a) = (-a) + a = 0$.

A group $(G, +)$ with the property that $a_1 + a_2 = a_2 + a_1$ for all $a_1$ and $a_2$ in $G$ is called Abelian (in honor of Niels Henrik Abel) or commutative. Groups not satisfying this property are said to be non-Abelian or non-commutative.

# Multiplicative Group

A group is a set, $G$, together with an operation $\times$ (called the group law of $G$) that combines any two elements $a_1$ and $a_2$ to form another element, denoted $a_1 \times a_2$. To qualify as a group, the set and operation, $(G, \times)$, must satisfy four requirements known as the group axioms:

(1) Closure: For all $a_1$ and $a_2$ in $G$, the result of the operation, $a_1 \times a_2$, is also in $G$.

(2) Associativity: For all $a_1$, $a_2$ and $a_3$ in $G$, $(a_1 \times a_2) \times a_3 = a_1 \times (a_2 \times a_3)$.

(3) Identity element: There exists an element 1 in $G$ such that, for every element $a$ in $G$, the equation $1 \times a = a \times 1 = a$ holds. Such an element is unique, and thus one speaks of the identity element.

(4) Multiplicative Inverse: For each $a$ in $G$, there exists an inverse element $a^{-1}$ in $G$ such that $a \times a^{-1} = a^{-1} \times a = 1$.

A group $(G, \times)$ with the property that $a_1 \times a_2 = a_2 \times a_1$ for all $a_1$ and $a_2$ in $G$ is called Abelian (in honor of Niels Henrik Abel) or commutative. Groups not satisfying this property are said to be non-Abelian or non-commutative.

# Group

A group is a set, $G$, together with an operation $\cdot$ (called the group law of $G$) that combines any two elements $a_1$ and $a_2$ to form another element, denoted $a_1 \cdot a_2$. To qualify as a group, the set and operation, $(G, \cdot)$, must satisfy four requirements known as the group axioms:

(1) Closure: For all $a_1$ and $a_2$ in $G$, the result of the operation, $a_1 \cdot a_2$, is also in $G$.

(2) Associativity: For all $a_1$, $a_2$ and $a_3$ in $G$, $(a_1 \cdot a_2) \cdot a_3 = a_1 \cdot (a_2 \cdot a_3)$.

(3) Identity element: There exists an element $e$ in $G$ such that, for every element $a$ in $G$, the equation $e \cdot a = a \cdot e = a$ holds. Such an element is unique, and thus one speaks of the identity element.

(4) Inverse element: For each $a$ in $G$, there exists an inverse element $a^{-1}$ in $G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

A group $(G, \cdot)$ with the property that $a_1 \cdot a_2 = a_2 \cdot a_1$ for all $a_1$ and $a_2$ in $G$ is called Abelian (in honor of Niels Henrik Abel) or commutative. Groups not satisfying this property are said to be non-Abelian or non-commutative.

# Ring

A ring is a set $R$ equipped with two binary operations $+$ and $\cdot$ satisfying the following three sets of axioms, called the ring axioms.

(1) $R$ is an abelian group under addition $+$, meaning that:

(1.1) Commutativity: For all $a_1$ and $a_2$ in $R$, the result of the operation, $a_1 + a_2 = a_2 + a_1$, is also in $R$.

(1.2) Associativity: For all $a_1$, $a_2$ and $a_3$ in $R$, $(a_1 + a_2) + a_3 = a_1 + (a_2 + a_3)$.

(1.3) Additive identity: There exists an element 0 in $R$ such that, for every element $a$ in $R$, the equation $0 + a = a + 0 = a$ holds.

(1.4) Additive Inverse: For each $a$ in $R$, there exists an inverse element $-a$ in $R$ such that $a + (-a) = (-a) + a = 0$.

(2) $R$ is a monoid under multiplication $\times$, meaning that:

(2.1) Closure: For all $a_1$ and $a_2$ in $R$, the result of the operation, $a_1 \times a_2$, is also in $R$.

(2.2) Associativity: For all $a_1$, $a_2$ and $a_3$ in $R$, $(a_1 \times a_2) \times a_3 = a_1 \times (a_2 \times a_3)$.

(2.3) Multiplicative Identity: There exists an element 1 in $R$ such that, for every element $a$ in $R$, the equation $1 \times a = a \times 1 = a$ holds.

(3) Multiplication is distributive with respect to addition:

(3.1) Left Distributivity: For all $a_1$, $a_2$ and $a_3$ in $R$, $a_1 \times (a_2 + a_3) = a_1 \times a_2 + a_1 \times a_3$.

(3.2) Right Distributivity: For all $a_1$, $a_2$ and $a_3$ in $R$, $(a_2 + a_3) \times a_1 = a_2 \times a_1 + a_3 \times a_1$.

# Abelian Ring = Commutative Ring

A ring is a set $R$ equipped with two binary operations $+$ and $\cdot$ satisfying the following three sets of axioms, called the ring axioms.

(1) $R$ is an abelian group under addition $+$, meaning that:

(1.1) Commutativity: For all $a_1$ and $a_2$ in $R$, the result of the operation, $a_1 + a_2 = a_2 + a_1$, is also in $R$.

(1.2) Associativity: For all $a_1$, $a_2$ and $a_3$ in $R$, $(a_1 + a_2) + a_3 = a_1 + (a_2 + a_3)$.

(1.3) Additive identity: There exists an element 0 in $R$ such that, for every element $a$ in $R$, the equation $0 + a = a + 0 = a$ holds.

(1.4) Additive Inverse: For each $a$ in $R$, there exists an inverse element $-a$ in $R$ such that $a + (-a) = (-a) + a = 0$.

(2) $R$ is a monoid under multiplication $\times$, meaning that:

(2.1) Commutativity: For all $a_1$ and $a_2$ in $R$, the result of the operation, $a_1 \times a_2 = a_2 \times a_1$, is also in $R$.

(2.2) Associativity: For all $a_1$, $a_2$ and $a_3$ in $R$, $(a_1 \times a_2) \times a_3 = a_1 \times (a_2 \times a_3)$.

(2.3) Multiplicative Identity: There exists an element 1 in $R$ such that, for every element $a$ in $R$, the equation $1 \times a = a \times 1 = a$ holds.

(3) Multiplication is distributive with respect to addition:

(3.1) Left Distributivity: For all $a_1$, $a_2$ and $a_3$ in $R$, $a_1 \times (a_2 + a_3) = a_1 \times a_2 + a_1 \times a_3$.

(3.2) Right Distributivity: For all $a_1$, $a_2$ and $a_3$ in $R$, $(a_2 + a_3) \times a_1 = a_2 \times a_1 + a_3 \times a_1$.

# Field

A field is a set $F$ equipped with two binary operations $+$ and $\cdot$ satisfying the following three sets of axioms, called the ring axioms.

(1) $F$ is an abelian group under addition $+$, meaning that:

(1.1) Commutativity: For all $a_1$ and $a_2$ in $F$, the result of the operation, $a_1 + a_2 = a_2 + a_1$, is also in $F$.

(1.2) Associativity: For all $a_1$, $a_2$ and $a_3$ in $F$, $(a_1 + a_2) + a_3 = a_1 + (a_2 + a_3)$.

(1.3) Additive identity: There exists an element 0 in $F$ such that, for every element $a$ in $F$, the equation $0 + a = a + 0 = a$ holds.

(1.4) Additive Inverse: For each $a$ in $F$, there exists an inverse element $-a$ in $F$ such that $a + (-a) = (-a) + a = 0$.

(2) $F$ is a monoid under multiplication $\times$, meaning that:

(2.1) Commutativity: For all $a_1$ and $a_2$ in $F$, the result of the operation, $a_1 \times a_2 = a_2 \times a_1$, is also in $F$.

(2.2) Associativity: For all $a_1$, $a_2$ and $a_3$ in $F$, $(a_1 \times a_2) \times a_3 = a_1 \times (a_2 \times a_3)$.

(2.3) Multiplicative Identity: There exists an element 1 in $F$ such that, for every element $a$ in $F$, the equation $1 \times a = a \times 1 = a$ holds.

(2.4) Multiplicative Inverse: For each $a \neq 0$ in $F$, there exists an inverse element $a^{-1}$ in $F$ such that $a \times a^{-1} = a^{-1} \times a = 1$.

(3) Multiplication is distributive with respect to addition:

(3.1) Left Distributivity: For all $a_1$, $a_2$ and $a_3$ in $F$, $a_1 \times (a_2 + a_3) = a_1 \times a_2 + a_1 \times a_3$.

(3.2) Right Distributivity: For all $a_1$, $a_2$ and $a_3$ in $R$, $(a_2 + a_3) \times a_1 = a_2 \times a_1 + a_3 \times a_1$.

# Field

If there is an element 1 in $R$ such that $1 \neq 0$ and $1 \times a = a \times 1 = a$ for each element $a$ in $R$, we say that $R$ is a ring with unity or identity. A commutative ring $R$ with identity is called an integral domain if, for every $a_1$ and $a_2$ in $R$ such that $a_1 \times a_2 = 0$, either $a_1 = 0$ or $a_2 = 0$. A division ring is a ring $R$, with an identity, in which every non-zero element in $R$ is a unit; that is, for each $a$ in $R$ with $a \neq 0$, there exists a unique element $a^{-1}$ such that $a^{-1} \times a = a \times a^{-1} = 1$. A commutative division ring is called a field.

Rings: $\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$

Fields: $\mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$

# Linear Space = Vector Space

A vector space over a field $F$ is a set $V$ together with two operations that satisfy the eight axioms listed below.

(1) The first operation, called vector addition or simply addition $+ : V \times V \to V$, takes any two vectors $v_1$ and $v_2$ and assigns to them a third vector which is commonly written as $v_1 + v_2$, and called the sum of these two vectors. (Note that the resultant vector is also an element of the set $V$).

(2) The second operation, called scalar multiplication $\cdot : F \times V \to V$, takes any scalar $a$ and any vector $v$ and gives another vector $a \cdot v$. (Similarly, the vector $a \cdot v$ is an element of the set $V$).

Elements of $V$ are commonly called vectors. Elements of $F$ are commonly called scalars.

# Linear Space = Vector Space

(1) Commutativity: For all $v_1$ and $v_2$ in $F$, the result of the operation, $v_1 + v_2 = v_2 + v_1$, is also in $V$.

(2) Associativity: For all $v_1$, $v_2$ and $v_3$ in $V$, $(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$.

(3) Additive identity: There exists an element 0 in $V$ such that, for every element $v$ in $V$, the equation $0 + v = v + 0 = v$ holds.

(4) Additive Inverse: For each $v$ in $V$, there exists an inverse element $-v$ in $V$ such that $v + (-v) = (-v) + v = 0$.

(5) Multiplicative Identity in Field $F$ for Scalar Multiplication: For each $v$ in $V$, $1 \cdot v = v$.

(6) Compatibility of Scalar Multiplication with Field Multiplication: For any $a_1$ and $a_2$ in $F$ and each $v$ in $V$, $a_1 \cdot (a_2 \cdot v) = (a_1 \times a_2) \cdot v$.

(7) Distributivity of Scalar Multiplication with respect to Vector Addition: For any $a$ in $F$ and each $v_1$ and $v_2$ in $V$, $a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$.

(8) Distributivity of Scalar Multiplication with respect to Field Addition: For any $a_1$ and $a_2$ in $F$ and each $v$ in $V$, $(a_1 + a_2) \cdot v = a_1 \cdot v + a_2 \cdot v$.

# Order Axioms

**Order Axioms** There is a subset $\mathbb{P}$ of the set $\mathbb{R}$ of real numbers, called the set of strictly positive numbers such that:

(1) For any real number $a$ in $\mathbb{R}$, exactly one of the following holds: $a = 0$ or $a \in \mathbb{P}$ or $-a \in \mathbb{P}$.

(2) If $a_1$ and $a_2$ in $\mathbb{P}$, $a_1 + a_2 \in \mathbb{P}$.

(3) If $a_1$ and $a_2$ in $\mathbb{P}$, $a_1 \cdot a_2 \in \mathbb{P}$.

**The "Less Than" Relation** We now define

- $a_1 < a_2$ to mean $a_2 - a_1 \in \mathbb{P}$ $(a_2 - a_1 \in \mathbb{R}_{++})$
- $a_1 \leqslant a_2$ to mean $a_2 - a_1 \in \mathbb{P}$ or $a_1 = a_2$ $(a_2 - a_1 \in \mathbb{R}_{+})$
- $a_1 > a_2$ to mean $a_1 - a_2 \in \mathbb{P}$ $(a_1 - a_2 \in \mathbb{R}_{++})$
- $a_1 \geqslant a_2$ to mean $a_1 - a_2 \in \mathbb{P}$ or $a_1 = a_2$ $(a_1 - a_2 \in \mathbb{R}_{+})$

$a_1 < a_2$ if and only if $a_2 > a_1$ and $a_1 \leqslant a_2$ if and only if $a_2 \geqslant a_1$.

$$\mathbb{P} = \mathbb{R}_+ \setminus \{0\} = \mathbb{R}_{++}$$

# Ordered Field

Any field $F$, together with two operations $\oplus$ and $\otimes$ and two members $0_{\oplus}$ and $0_{\otimes}$ of $F$, and a subset $P$ of $F$, which satisfies the corresponding versions of (1) - (2) - (3), is called an ordered field.

Example. Both $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields.

# Completeness Axiom

Dedekind Completeness Axiom Suppose $A$ and $B$ are two (non-empty) sets of real numbers with the properties:

(1) if $a \in A$ and $b \in B$, then $a < b$.

(2) every real number is in either $A$ or $B$ (in symbols; $A \cup B = \mathbb{R}$).

Then there is a unique real number $c$ such that:

(1) if $a < c$ then $a \in A$, and

(2) if $b > c$ then $b \in B$.

Note that every number strictly less than $c$ belongs to $A$ and every number strictly more than $c$ belongs to $B$. Moreover, either $c \in A$ or $c \in B$ by (2). Hence if $c \in A$ then $A = (-\infty, c]$ and $B = (c, \infty)$; while if $c \in B$ then $A = (-\infty, c)$ and $B = [c, \infty)$.

The pair of sets $\{A, B\}$ is called a Dedekind Cut.

The intuitive idea of Dedekind Completeness Axiom is that there are no "holes" in the real numbers.

# Dedekind - Cantor Axiom of Continuity of a Real Line

Dedekind - Cantor Axiom of Continuity of a Real Line Suppose the pair of sets $(A, A^c)$ is a Dedekind Cut of $\mathbb{R}$ with the property: if $a \in A$ and $b \in A^c$, then $a < b$. Then either there exists a maximum in $A$ or there exists a minimum in $A^c$:

(1) If there exists a maximum $c$ in $A$, then $A = (-\infty, c]$ and $A^c = (c, \infty)$.

(2) If there exists a minimum $c$ in $A^c$, then $A = (-\infty, c)$ and $A^c = [c, \infty)$.

# Supremum and Infimum

(1) Let $S$ be a set of real numbers. $u$ is an upper bound for $S$ if $x \leqslant u$ for all $x \in S$. A set $S$ is bounded above if it has an upper bound.

(2) $b$ is the least upper bound (or l.u.b. or supremum or sup) for $S$ if $b$ is an upper bound, and moreover $b \leqslant u$ whenever $u$ is any upper bound for $S$.

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(1) Let $S$ be a set of real numbers. $d$ is an lower bound for $S$ if $x \geqslant d$ for all $x \in S$. A set $S$ is bounded below if it has a lower bound.

(2) $l$ is the greatest lower bound (or g.l.b. or infimum or inf) for $S$ if $l$ is an lower bound, and moreover $l \geqslant d$ whenever $d$ is any lower bound for $S$.

# Supremum and Infimum Completeness Axiom

**Theorem of Supremum (Least Upper Bound Completeness Axiom)**
Suppose $S$ is a non-empty set of real numbers which is bounded above. Then $S$ has a supremum in $\mathbb{R}$.

**Theorem of Infimum (Greatest Lower Bound Completeness Axiom)**
Suppose $S$ is a non-empty set of real numbers which is bounded below. Then $S$ has a infimum in $\mathbb{R}$.

# Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Necassity] Suppose that $S$ is a non-empty set of real numbers which is bounded above. Let

$$B = \{x \in \mathbb{R} : x \text{ is an upper bound for } S\} \quad \text{and} \quad A = \mathbb{R} \setminus B.$$

Note that $B \neq \emptyset$; and if $x \in S$ then $x - 1$ is not an upper bound for $S$ so $A \neq \emptyset$.

Suppose $a \in A$ and $b \in B$, then $a < b$. If $a \geqslant b$, then $a$ would also be an upper bound for $S$, which contradicts the definition of $A$.

Let $c$ be the real number given by Dedekind - Cantor Axiom. We claim that $c = \sup S$.

If $c \in A$ then $c$ is not an upper bound for $S$ and so there exists $x \in S$ with $c < x$. But then $a = \dfrac{c + x}{2}$ is not an upper bound for $S$, i.e. $a \in A$, contradicting the fact from the conclusion of Dedekind - Cantor Axiom that $a \leqslant c$ for all $a \in A$. Hence $c \in B$. But if $c \in B$ then $c \leqslant b$ for all $b \in B$; i.e. $c$ is less than any upper bound for $S$.

# Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Sufficiency] For this, suppose $\{A, B\}$ is a Dedekind cut. Then $A$ is bounded above (by any element of $B$). Let $c = \sup A$. We claim that

$$a < c \Rightarrow a \in A, \qquad \text{and} \qquad b > c \Rightarrow b \in B.$$

- Suppose $a < c$. Now every member of $B$ is an upper bound for $A$, from the first property of a Dedekind cut; hence $a \notin B$, as otherwise $a$ would be an upper bound for $A$ which is less than the least upper bound $c$. Hence $a \in A$.
- Next suppose $b > c$. Since $c$ is an upper bound for $A$ (in fact the least upper bound), it follows we cannot have $b \in A$, and thus $b \in B$.

# Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Necassity] Suppose that $S$ is a non-empty set of real numbers which is bounded below. Let

$$A = \{x \in \mathbb{R} : x \text{ is a lower bound for } S\} \quad \text{and} \quad B = \mathbb{R} \setminus A.$$

Note that $A \neq \emptyset$; and if $x \in S$ then $x + 1$ is not a lower bound for $S$ so $B \neq \emptyset$.

Suppose $a \in A$ and $b \in B$, then $a < b$. If $a \geqslant b$, then $b$ would also be a lower bound for $S$, which contradicts the definition of $B$.

Let $c$ be the real number given by Dedekind - Cantor Axiom. We claim that $c = \inf S$.

If $c \in B$ then $c$ is not a lower bound for $S$ and so there exists $x \in S$ with $x < c$. But then $b = \dfrac{x + c}{2}$ is not a lower bound for $S$, i.e. $b \in B$, contradicting the fact from the conclusion of Dedekind - Cantor Axiom that $c \leqslant b$ for all $b \in B$. Hence $c \in A$. But if $c \in A$ then $a \leqslant c$ for all $a \in A$; i.e. $c$ is more than any lower bound for $S$.

# Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Sufficiency] For this, suppose $\{A, B\}$ is a Dedekind cut. Then $B$ is bounded below (by any element of $A$). Let $c = \inf B$. We claim that

$$a < c \Rightarrow a \in A, \qquad \text{and} \qquad b > c \Rightarrow b \in B.$$

- Suppose $a < c$. Since $c$ is a lower bound for $B$ (in fact the greatest lower bound), it follows we cannot have $a \in B$, and thus $a \in A$.
- Next suppose $c < b$. Now every member of $A$ is a lower bound for $B$, from the first property of a Dedekind cut; hence $b \notin A$, as otherwise $b$ would be a lower bound for $B$ which is less than the greatest lower bound $c$. Hence $b \in B$.

# Proposition on Supremum and Infimum

Suppose $S$ is a non-empty set of real numbers. Then $l = \inf S$ iff

(1) $x \geqslant l$ for all $x \in S$;

(2) for each $\varepsilon > 0$ there exists $x \in S$ such that $x < l + \varepsilon$.

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Suppose $S$ is a non-empty set of real numbers. Then $b = \sup S$ iff

(1) $x \leqslant b$ for all $x \in S$;

(2) for each $\varepsilon > 0$ there exists $x \in S$ such that $x > b - \varepsilon$.

# Definition of Supremum

Suppose $S$ is a non-empty set of real numbers. Then $b = \sup S$ iff

(1) $x \leqslant b$ for all $x \in S$;

(2) $b \leqslant u$ whenever $u$ is any upper bound for $S$.

Suppose $S$ is a non-empty set of real numbers. Then $b = \sup S$ iff

(1) $x \leqslant b$ for all $x \in S$;

(2′) for each $\varepsilon > 0$ there exists $x \in S$ such that $x > b - \varepsilon$.

$[(1) + (2) \Rightarrow (2')]$ Suppose $(2')$ is not true. Then for some $\varepsilon > 0$ it follows that $x \leqslant b - \varepsilon$ for every $x \in S$, i.e. $b - \varepsilon$ is an upper bound for $S$. This is a contradiction.

$[(1) + (2') \Rightarrow (2)]$ If $u < b$ then from $(2')$ it follows that $u$ is not an upper bound of $S$. That is (2): $b \leqslant u$ whenever $u$ is any upper bound for $S$.

# Definition of Infimum

Suppose $S$ is a non-empty set of real numbers. Then $l = \inf S$ iff

(1) $x \geqslant l$ for all $x \in S$;

(2) $l \geqslant d$ whenever $d$ is any lower bound for $S$.

Suppose $S$ is a non-empty set of real numbers. Then $l = \inf S$ iff

(1) $x \geqslant l$ for all $x \in S$;

(2′) for each $\varepsilon > 0$ there exists $x \in S$ such that $x < l + \varepsilon$.

$[(1) + (2) \Rightarrow (2')]$ Suppose $(2')$ is not true. Then for some $\varepsilon > 0$ it follows that $x \geqslant l + \varepsilon$ for every $x \in S$, i.e. $l + \varepsilon$ is a lower bound for $S$. This is a contradiction.

$[(1) + (2') \Rightarrow (2)]$ If $d > l$ then from $(2')$ it follows that $d$ is not a lower bound of $S$. That is (2): $l \geqslant d$ whenever $d$ is any lower bound for $S$.

# Bounded Monotone Sequence

Definition. A sequence is bounded if the set of terms from the sequence is bounded.

Definition. A sequence $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is

(1) increasing if $x_n \leqslant x_{n+1}$ for all $n \in \mathbb{N}$;

(2) decreasing if $x_n \geqslant x_{n+1}$ for all $n \in \mathbb{N}$;

(3) strictly increasing if $x_n < x_{n+1}$ for all $n \in \mathbb{N}$;

(4) strictly decreasing if $x_n > x_{n+1}$ for all $n \in \mathbb{N}$.

A sequence is monotone if it is either increasing or decreasing.

# Theorem on Limit of Bounded Monotone Sequence

**Theorem on Limit of Bounded Monotone Sequence** Every bounded monotone sequence in $\mathbb{R}$ has a limit in $\mathbb{R}$.

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**Theorem of Limit of Increasing Sequence** Suppose $\{x_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

**Theorem of Limit of Decreasing Sequence** Suppose $\{x_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

# Theorem on Limit of Bounded Monotone Sequence

**Theorem of Limit of Increasing Sequence** Suppose $\{x_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

**Proof.** Suppose $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ and $\{x_n : n \in \mathbb{N}\}$ is increasing. Since the set of terms $\{x_n : n \in \mathbb{N}\}$ is bounded above, it has a supremum $x$. We claim that $\lim_{n\to\infty} x_n = x$.

To see this, note that $x_n \leqslant x$ for all $n \in \mathbb{N}$; but if $\varepsilon > 0$ then there exists $N \in \mathbb{N}$ such that $x_N > x - \varepsilon$, as otherwise $x - \varepsilon$ would be an upper bound. Choose such $N = N(\varepsilon)$. Since $x_N > x - \varepsilon$, then $x_n > x - \varepsilon$ for all $n \geqslant N$ as the sequence is increasing. Hence

$$x - \varepsilon < x_n \leqslant x$$

for all $n \geqslant N$. Thus $|x - x_n| < \varepsilon$ for $n \geqslant N$, and so $\lim_{n\to\infty} x_n = x$ (since $\varepsilon > 0$ is arbitrary).

# Theorem on Limit of Bounded Monotone Sequence

**Theorem of Limit of Decreasing Sequence** Suppose $\{x_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

*Proof.* Suppose $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ and $\{x_n : n \in \mathbb{N}\}$ is decreasing. Since the set of terms $\{x_n : n \in \mathbb{N}\}$ is bounded below, it has a supremum $x$. We claim that $\lim_{n\to\infty} x_n = x$.

To see this, note that $x_n \geqslant x$ for all $n \in \mathbb{N}$; but if $\varepsilon > 0$ then there exists $N \in \mathbb{N}$ such that $x_N < x + \varepsilon$, as otherwise $x + \varepsilon$ would be a lower bound. Choose such $N = N(\varepsilon)$. Since $x_N < x + \varepsilon$, then $x_n < x + \varepsilon$ for all $n \geqslant N$ as the sequence is increasing. Hence

$$x \leqslant x_n < x + \varepsilon$$

for all $n \geqslant N$. Thus $|x - x_n| < \varepsilon$ for $n \geqslant N$, and so $\lim_{n\to\infty} x_n = x$ (since $\varepsilon > 0$ is arbitrary).

# Cantor Nested Intervals Theorem

**Theorem 1** If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $D = sup\{d_n : n \in \mathbb{N}\}$ such that $D \in \bigcap_{n=1}^{\infty} I_n$.

**Theorem 2** If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $U = inf\{u_n : n \in \mathbb{N}\}$ such that $U \in \bigcap_{n=1}^{\infty} I_n$.

**Nested Intervals Theorem** If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exist two real numbers $D = sup\{d_n : n \in \mathbb{N}\}$ and $U = inf\{u_n : n \in \mathbb{N}\}$ such that $[D, U] = \bigcap_{n=1}^{\infty} I_n$.

**Nested Intervals Theorem** If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals and $\lim_{n\to\infty} |I_n| = \lim_{n\to\infty} (u_n - d_n) = 0$, then

$$\lim_{n\to\infty} d_n = \lim_{n\to\infty} u_n \in \bigcap_{n=1}^{\infty} I_n.$$

$\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals if $I_1 \supseteq I_2 \supseteq \cdots \supseteq I_n \supseteq \cdots$.

# Cantor Nested Intervals Theorem

**Theorem 1** If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $D = sup\{d_n : n \in \mathbb{N}\}$ such that $D \in \bigcap_{n=1}^{\infty} I_n$.

*Proof.* We note that by the definition of nested intervals that $I_n \subseteq I_1$ for all $n \in \mathbb{N}$ so then $d_n \leqslant u_1$ for all $n \in \mathbb{N}$.
Now consider the non-empty set $A = \{d_n : n \in \mathbb{N}\}$ that is bounded above by $u_1$. Thus this set has a supremum in the real numbers and denote it $D = \sup A = \lim_{n\to\infty} d_n$ so that $d_n \leqslant D$ for all $n \in \mathbb{N}$.
We now want to show that $D \leqslant u_n$ for all $n \in \mathbb{N}$. We will do this by showing that $d_n \leqslant u_{n'}$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$.
First consider the case where $n \leqslant n'$. We thus have that $I_n \supseteq I_{n'}$ by the definition of nested intervals and so we get that $d_n \leqslant d_{n'} \leqslant u_{n'} \leqslant u_n$. Here we see that $d_n \leqslant u_{n'}$.
Now consider the case where $n > n'$. We thus have that $I_{n'} \supseteq I_n$ by the definition of nested intervals and so we have that $d_{n'} \leqslant d_n \leqslant u_n \leqslant u_{n'}$. Once again we have that $d_n \leqslant u_{n'}$.
So then $d_n \leqslant u_{n'}$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$, and so $u_{n'}$ is an upper bound to the set $A$ and so $D = \sup A \leqslant u_{n'}$ for all $n' \in \mathbb{N}$. Furthermore we have that $d_{n'} \leqslant D$ for all $n' \in \mathbb{N}$, and so $D \in I_{n'}$ for every $n' \in \mathbb{N}$ and thus $D \in \bigcap_{n=1}^{\infty} I_n$ and so the set theoretic union is non-empty.

# Cantor Nested Intervals Theorem

Theorem 2 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $U = inf\{u_n : n \in \mathbb{N}\}$ such that $U \in \bigcap_{n=1}^{\infty} I_n$.

Proof. We note that by the definition of nested intervals that $I_n \subseteq I_1$ for all $n \in \mathbb{N}$ so then $d_1 \leqslant u_n$ for all $n \in \mathbb{N}$.
Now consider the non-empty set $B = \{u_n : n \in \mathbb{N}\}$ that is bounded above by $d_1$. Thus this set has an infumum in the real numbers and denote it $U = \inf B = \lim_{n\to\infty} u_n$ so that $U \leqslant u_n$ for all $n \in \mathbb{N}$.
We now want to show that $l_n \leqslant U$ for all $n \in \mathbb{N}$. We will do this by showing that $d_{n'} \leqslant u_n$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$.
First consider the case where $n \leqslant n'$. We thus have that $I_n \supseteq I_{n'}$ by the definition of nested intervals and so we get that $d_n \leqslant d_{n'} \leqslant u_{n'} \leqslant u_n$. Here we see that $d_{n'} \leqslant u_n$.
Now consider the case where $n > n'$. We thus have that $I_{n'} \supseteq I_n$ by the definition of nested intervals and so we have that $d_{n'} \leqslant d_n \leqslant u_n \leqslant u_{n'}$. Once again we have that $d_{n'} \leqslant u_n$.
So then $d_{n'} \leqslant u_n$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$, and so $l_{n'}$ is a lower bound to the set $B$ and so $u_{n'} \leqslant \sup B = U$ for all $n' \in \mathbb{N}$. Furthermore we have that $U \leqslant u_{n'}$ for all $n' \in \mathbb{N}$, and so $U \in I_{n'}$ for every $n' \in \mathbb{N}$ and thus $U \in \bigcap_{n=1}^{\infty} I_n$ and so the set theoretic union is non-empty.

# Cantor Nested Intervals Theorem

**Nested Intervals Theorem** If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exist two real numbers $D = sup\{d_n : n \in \mathbb{N}\} = \lim_{n\to\infty} d_n$ and $U = inf\{u_n : n \in \mathbb{N}\} = \lim_{n\to\infty} u_n$ such that $[D, U] = \bigcap_{n=1}^{\infty} I_n$.

**Proof.** We first prove that $[D, U] \subseteq \bigcap_{n=1}^{\infty} I_n$. Now let $x \in [D, U]$, then $D \leqslant x \leqslant U$. But we know that $d_n \leqslant D$ for all $n \in \mathbb{N}$ and $U \leqslant u_n$ for all $n \in \mathbb{N}$ and so $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$. Therefore $x \in [d_n, u_n]$ for all $n \in \mathbb{N}$ or in other words $x \in \bigcap_{n=1}^{\infty} I_n$.

Therefore $[D, U] \subseteq \bigcap_{n=1}^{\infty} I_n$.

We then prove that $\bigcap_{n=1}^{\infty} I_n \subseteq [D, U]$. Let $x \in \bigcap_{n=1}^{\infty} I_n$. Then $d_n \leqslant x \leqslant u_n$ for all $n \in \mathbb{N}$. We also know that $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$. Suppose that $x$ is such that $d_n \leqslant x < D$. Since $D$ is the supremum of the set $A$ then there exists an element $d_x \in \{d_n : n \in \mathbb{N}\}$ such that $x < d_x$ and so $x \notin [d_x, u_x]$ and therefore $x \notin \bigcap_{n=1}^{\infty} I_n$, a contradiction. Now suppose that $x$ is such that $U < x \leqslant u_n$ for all $n \in \mathbb{N}$. Since $U$ is the infimum of the set $B$ then there exists an element $u_x \in \{u_n : n \in \mathbb{N}\}$ such that $u_x < x$ and so $x \notin [d_x, u_x]$ and so $x \notin \bigcap_{n=1}^{\infty} I_n$, once again, a contradiction. We must therefore have that $d_n \leqslant D \leqslant x \leqslant U \leqslant u_n$ and so $x \in [D, U]$ and so $\bigcap_{n=1}^{\infty} I_n \subseteq [D, U]$.

# Cantor Nested Intervals Theorem

**Nested Intervals Theorem** If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals and $\lim_{n\to\infty} |I_n| = \lim_{n\to\infty} (u_n - d_n) = 0$, then $\lim_{n\to\infty} d_n = \lim_{n\to\infty} u_n \in \bigcap_{n=1}^{\infty} I_n$.

**Proof.** The non-empty set $A = \{d_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above by $u_1$, then it has a limit $D = \lim_{n\to\infty} d_n$.
The non-empty set $B = \{u_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below by $d_1$, then it has a limit $U = \lim_{n\to\infty} u_n$.
Thus $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$, and

$$0 \leqslant U - D \leqslant u_n - d_n.$$

Note $\lim_{n\to\infty} |I_n| = \lim_{n\to\infty} (u_n - d_n) = 0$, then $D = U \in \bigcap_{n=1}^{\infty} I_n$.

# Bolzano Theorem

Bolzano Theorem Each bounded sequence in $\mathbb{R}$ has a convergent subsequence.
Bolzano Theorem Each bounded sequence in $\mathbb{R}$ has a limit point.

Proof. Let $\{x_n : n \in \mathbb{N}\}$ be a bounded sequence, we assume $\{x_n : n \in \mathbb{N}\} \subseteq [D, U]$. We divide closed interval $[D, U]$ into two equal-length parts $\left[D, \frac{D+U}{2}\right]$ and $\left[\frac{D+U}{2}, U\right]$. There exists an interval including infinite items of sequence $\{x_n : n \in \mathbb{N}\}$, this interval is written as $[D_1, U_1]$, and we take $x_{n_1} \in [D_1, U_1]$. Then we divide closed interval $[D_1, U_1]$ into two equal-length parts $\left[D_1, \frac{D_1+U_1}{2}\right]$ and $\left[\frac{D_1+U_1}{2}, U_1\right]$. There exists an interval including infinite items of sequence $\{x_n : n \in \mathbb{N}\}$, this interval is written as $[D_2, U_2]$, and we take $x_{n_2} \in [D_2, U_2]$ with $n_2 > n_1$. We repeat this step and obtain a sequence $\{[D_k, U_k] : k \in \mathbb{N}\}$ of closed bounded nested intervals:

$$[D_1, U_1] \supseteq [D_2, U_2] \supseteq \cdots \supseteq [D_k, U_k] \supseteq \cdots \quad \text{with } U_k - D_k = \frac{U - D}{2^k}$$

with a subsequence $\{x_{n_k} : k \in \mathbb{N}\}$ with $x_{n_k} \in [D_k, U_k]$ for $k \in \mathbb{N}$.

From Cantor Nested Intervals Theorem there exists a unique point $a \in \bigcap_{k=1}^{\infty} [D_k, U_k]$, and $|x_{n_k} - a| < U_k - D_k = \frac{U-D}{2^k}$, then $\lim_{k\to\infty} x_{n_k} = a$.

# Weierstrass Theorem on Accumulation Point

Weierstrass Theorem Every bounded infinite set in $\mathbb{R}$ has an accumulation point.

Proof. Note that $E$ is a bounded infinite set in $\mathbb{R}$, there exists a constant $M$ such that, for any $x \in E$, $|x| \leqslant M$. Define $I_0 = [-M, M]$, we divide $R_0$ into two small closed intervals: $[-M, 0]$ and $[0, M]$, then at least one small interval includes infinite points in $E$ (otherwise $E$ is a finite set!), we call such a small closed interval $I_1$ with length $M$. Generally, we assume that there exists a sequence of closed intervals

$$I_0 \supseteq I_1 \supseteq \cdots \supseteq I_N, \quad I_N \cap E \text{ is an infinite set}$$

where the length of $I_n$ is $\frac{M}{2^{n-1}}$ for $n = 1, 2, \cdots, N$. We divide $I_N$ into two small closed intervals, then at least one small closed interval includes infinite points in $E$ (otherwise $E$ is a finite set!), we call such a small closed interval $I_{N+1}$ with length $\frac{M}{2^N}$. Thus we obtain a sequence of decreasing closed intervals

$$I_0 \supseteq I_1 \supseteq \cdots \supseteq I_n \supseteq \cdots, \quad I_n \cap E \text{ is an infinite set}$$

where the length of $I_n$ is $\frac{M}{2^{n-1}}$ for $n = 1, 2, \cdots$.

# Weierstrass Theorem on Accumulation Point

Weierstrass Theorem Every bounded infinite set in $\mathbb{R}$ has an accumulation point.

Note $\lim_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then, from Cantor Nested Intervals Theorem, there exists a unique point $x \in \bigcap_{n=1}^{\infty} I_n$. We now check that this $x$ is one of accumulation points of $E$.

Note $\lim_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then for enough large integer $n_0$, $\frac{M}{2^{n_0}} < \delta$. The length of $I_{n_0+1}$ is strictly less than $\delta$, then $I_{n_0+1} \subseteq N(x, \delta)$, and hence $E \cap I_{n_0+1} \subseteq E \cap N(x, \delta)$. $x$ is an accumulation point of $E$.

# Cauchy Convergence Criterion

Definition. $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is a Cauchy sequence if for any $\varepsilon > 0$ there exists an integer $N$ such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$.

Cauchy Convergence Criterion The sequence $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is convergent if and only if it is a Cauchy sequence.

# Cauchy Convergence Criterion

**Cauchy Convergence Criterion** The sequence $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is convergent if and only if it is a Cauchy sequence.

**Proof** [Necessity] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a convergent sequence, and suppose $x = \lim_{n\to\infty} x_n$. For any $\varepsilon > 0$ there exists an integer $N \in \mathbb{N}$ such that, when $n > N$, $|x_n - x| < \varepsilon$. It follows that for any $n_1 > N$ and $n_2 > N$,

$$|x_{n_1} - x_{n_2}| \leqslant |x_{n_1} - x| + |x_{n_2} - x| < 2\varepsilon.$$

Thus $\{x_n : n \in \mathbb{N}\}$ is a Cauchy sequence.

# Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer $N$ such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|x_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

From Bolzano-Weierstrass Theorem, the bounded sequence $\{x_n : n \in \mathbb{N}\}$ has a convergent subsequence $\{x_{n_k} : k \in \mathbb{N}\}$ with $\lim_{k\to\infty} x_{n_k} = a$. For any $\varepsilon > 0$ there exists an integer $K$ such that, when $k > K$, $|x_{n_k} - a| < \varepsilon$.

Take $k_0 = \max\{K+1, N+1\}$, then $n_{k_0} \geqslant n_{N+1} \geqslant N+1 > N$. Then, when $n > N$, $|x_n - x_{n_{k_0}}| < \varepsilon$, and hence

$$|x_n - a| \leqslant |x_n - x_{n_{k_0}}| + |x_{n_{k_0}} - a| < 2\varepsilon$$

that is, $\lim x_n = a$.

# Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer $N$ such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|n_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

Define $y_n = \inf\{x_n, x_{n+1}, \cdots\}$ for each $n \in \mathbb{N}$. It follows that $y_{n+1} \geqslant y_n$ since $y_{n+1}$ is the infimum over a subset of the set corresponding to $y_n$, $\{y_n : n \in \mathbb{N}\}$ is an increasing sequence. Moreover the sequence $\{y_n : n \in \mathbb{N}\}$ is bounded since the sequence $\{x_n : n \in \mathbb{N}\}$ is Cauchy and hence bounded. From Theorem on monotone sequences, there exists $a \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} y_n = a$. For any $\varepsilon > 0$ there exists an integer $N_1$ such that, when $n > N_1$, $|y_n - a| < \frac{1}{2}\varepsilon$, or $a - \frac{1}{2}\varepsilon < y_n < a + \frac{1}{2}\varepsilon$. We claim that $\lim\limits_{n\to\infty} x_n = a$.

# Cauchy Convergence Criterion

Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer $N_2$ such that, when $n_1 > N_2$ and $n_2 > N_2$, $|x_{n_1} - x_{n_2}| < \frac{1}{2}\varepsilon$, or

$$x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon.$$

From the definition of sequence $\{y_n : n \in \mathbb{N}\}$, $y_n \leqslant x_n$ for each $n \in \mathbb{N}$, then $y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2}$ for $n_1 > N_1$ and $n_2 > N_2$. On the other hand, $x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon$, then
$x_{n_2} \leqslant \inf\{x_{n_1} + \frac{1}{2}\varepsilon : n_1 \geqslant N_2\} = \inf\{x_{n_1} : n_1 \geqslant N_2\} + \frac{1}{2}\varepsilon = y_{n_1} + \frac{1}{2}\varepsilon$.
Thus there exists an integer $N_2$ such that, when $n_1 > N_2$ and $n_2 > N_2$,

$$y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_2} < y_{n_1} + \frac{1}{2}\varepsilon.$$

Take $N = \max\{N_1, N_2\}$, then, when $n_1 > N$ and $n_2 > N$,

$$a - \varepsilon < y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_2} < y_{n_1} + \frac{1}{2}\varepsilon < a + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, it follows that $\lim\limits_{n\to\infty} x_n = a$.

# Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer $N$ such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|n_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

Define $y_n = \sup\{x_n, x_{n+1}, \cdots\}$ for each $n \in \mathbb{N}$. It follows that $y_{n+1} \leqslant y_n$ since $y_{n+1}$ is the supremum over a subset of the set corresponding to $y_n$, $\{y_n : n \in \mathbb{N}\}$ is a decreasing sequence. Moreover the sequence $\{y_n : n \in \mathbb{N}\}$ is bounded since the sequence $\{x_n : n \in \mathbb{N}\}$ is Cauchy and hence bounded. From Theorem on monotone sequences, there exists $a \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} y_n = a$. For any $\varepsilon > 0$ there exists an integer $N_1$ such that, when $n > N_1$, $|y_n - a| < \frac{1}{2}\varepsilon$, or $a - \frac{1}{2}\varepsilon < y_n < a + \frac{1}{2}\varepsilon$. We claim that $\lim\limits_{n\to\infty} x_n = a$.

# Cauchy Convergence Criterion

Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer $N_2$ such that, when $n_1 > N_2$ and $n_2 > N_2$, $|x_{n_1} - x_{n_2}| < \frac{1}{2}\varepsilon$, or

$$x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon.$$

From the definition of sequence $\{y_n : n \in \mathbb{N}\}$, $y_n \geqslant x_n$ for each $n \in \mathbb{N}$, then $x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon \leqslant y_{n_1} + \frac{1}{2}\varepsilon$ for $n_1 > N_1$ and $n_2 > N_2$. On the other hand, $x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2}$, then
$y_{n_1} - \frac{1}{2}\varepsilon = \sup\{x_{n_1} : n_1 \geqslant N_2\} - \frac{1}{2}\varepsilon = \sup\{x_{n_1} - \frac{1}{2}\varepsilon : n_1 \geqslant N_2\} \leqslant x_{n_2}$. Thus there exists an integer $N_2$ such that, when $n_1 > N_2$ and $n_2 > N_2$,

$$y_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} \leqslant y_{n_1} + \frac{1}{2}\varepsilon.$$

Take $N = \max\{N_1, N_2\}$, then, when $n_1 > N$ and $n_2 > N$,

$$a - \varepsilon < y_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} \leqslant y_{n_1} + \frac{1}{2}\varepsilon < a + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, it follows that $\lim\limits_{n\to\infty} x_n = a$.

# Heine - Borel Theorem on Finite Coverings

A collection $\mathscr{H}$ of open sets is an open covering of a set $S$ if every point in $S$ is contained in a set $H$ belonging to $\mathscr{H}$; that is, $S \subseteq \{H \in \mathscr{H}\}$.

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset $S$ of the real line; then $S$ has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

# Heine - Borel Theorem on Finite Coverings

**Heine - Borel Theorem** If $\mathscr{H}$ is an open covering of a closed and bounded subset $S$ of the real line; then $S$ has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

**Proof. (Reduction to Absurdity)** Let $\mathscr{H}$ be an open covering of $[D, U]$. Assume that $[D, U]$ can not be covered by finite many open sets belonging to $\mathscr{H}$. We divide closed interval $[D, U]$ into two equal-length parts $\left[D, \frac{D+U}{2}\right]$ and $\left[\frac{D+U}{2}, U\right]$. There exists at least one small interval that can not be covered by finite many open sets belonging to $\mathscr{H}$, this interval is written as $[D_1, U_1]$. Then we divide closed interval $[D_1, U_1]$ into two equal-length parts, then there exists at least one small interval that can not be covered by finite many open sets belonging to $\mathscr{H}$, this interval is written as $[D_2, U_2]$. We repeat this step and obtain a sequence $\{[D_n, U_n] : n \in \mathbb{N}\}$ of closed bounded nested intervals that can not be covered by finite many open sets belonging to $\mathscr{H}$:

$$[D_1, U_1] \supseteq [D_2, U_2] \supseteq \cdots \supseteq [D_n, U_n] \supseteq \cdots \quad \text{with } U_n - D_n = \frac{U - D}{2^n}.$$

# Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset $S$ of the real line; then $S$ has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

From Cantor Nested Intervals Theorem there exists a unique point $x \in \bigcap_{n=1}^{\infty} [D_n, U_n]$. Note that $\mathscr{H}$ is an open covering of $[D, U]$, then there exists $H_0 \in \mathscr{H}$ such that $x \in H_0$. Note that $H_0$ is an open set, then there exists $\delta > 0$ such that $(x - \delta, x + \delta) \subseteq H_0$.

Note $\lim_{n\to\infty} D_n = x = \lim_{n\to\infty} U_n$, then there exists an integer $N$ such that, when $n > N$, $D_n \in (x - \delta, x + \delta)$ and $U_n \in (x - \delta, x + \delta)$.

$$[D_n, U_n] \subseteq (x - \delta, x + \delta) \subseteq H_0 \qquad \text{for } n > N.$$

which contradicts that $[D_n, U_n]$ can not be covered by finite many open sets belonging to $\mathscr{H}$.

# Real Number Space $\mathbb{R}$

Dedekind - Cantor Axiom of Continuity of a Real Line Suppose the pair of sets $(A, A^c)$ is a Dedekind Cut of $\mathbb{R}$ with the property: if $a \in A$ and $b \in A^c$, then $a < b$, then either there exists a maximum in $A$ or there exists a minimum in $A^c$.

Theorem of Supremum and Infimum Suppose $E$ is a non-empty set of real numbers which is bounded above, then $E$ has a supremum in $\mathbb{R}$. Suppose $E$ is a non-empty set of real numbers which is bounded below, then $E$ has a infimum in $\mathbb{R}$.

Theorem of Limit of Monotone Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above, then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists. Suppose $\{x_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below, then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Nested Intervals Theorem If the interval $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exist two real numbers $D = sup\{d_n : n \in \mathbb{N}\}$ and $U = inf\{u_n : n \in \mathbb{N}\}$ such that $[D, U] \subset \cap_{n=1}^{\infty} I_n$.

# Real Number Space $\mathbb{R}$

Bolzano Theorem Each bounded sequence in $\mathbb{R}$ has a convergent subsequence.

Weierstrass Theorem Every bounded infinite set in $\mathbb{R}$ has an accumulation point.

Cauchy Convergence Criterion The sequence $\{x_n : n \in \mathbb{N}\}$ in $\mathbb{R}$ is convergent if and only if it is a Cauchy sequence: for any $\varepsilon > 0$, there exist a large natural number $N$ such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$.

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset $E$ of the real line; then $E$ has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.